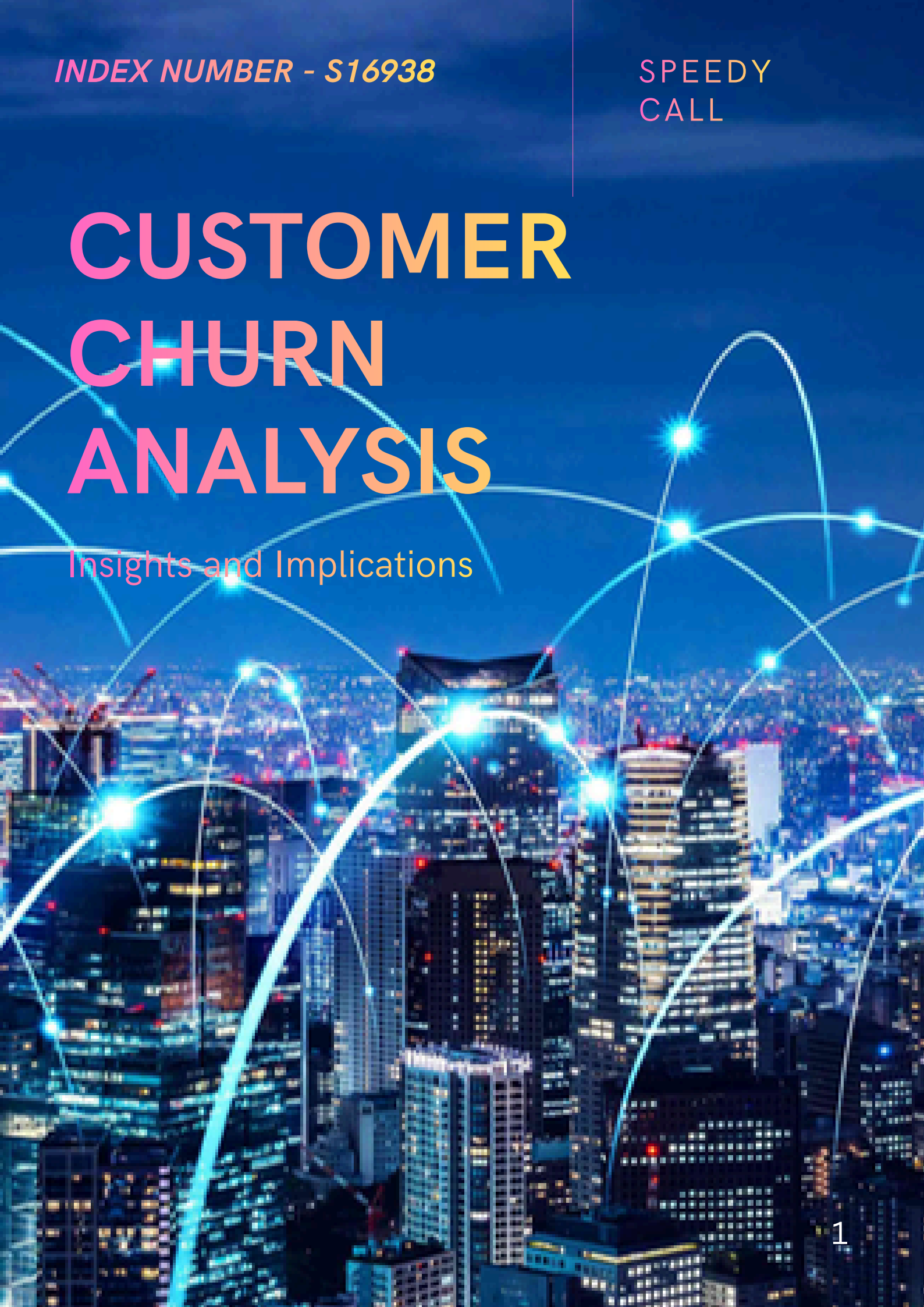
INDEX NUMBER - S16938

SPEEDY CALL

CUSTOMER CHURN ANALYSIS

Insights and Implications

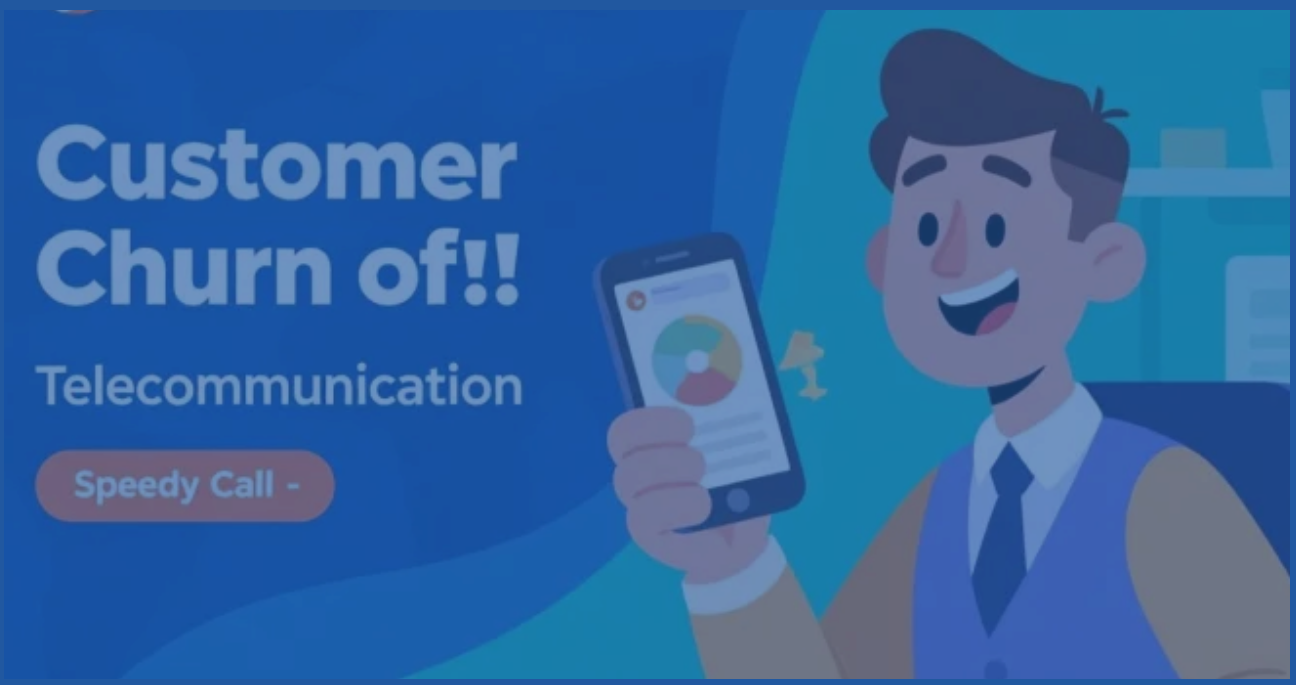


ABSTRACT

Customer churn is a critical challenge for telecom companies, as losing subscribers directly impacts revenue and growth. This study investigates the demographic, financial, and service-related factors that influence churn behavior. To begin, an exploratory analysis was carried out using bar charts, pie charts, histograms, and box plots to visualize customer characteristics and churn patterns. These plots provided initial insights into variable distributions and highlighted potential associations worth testing.

To formally assess these relationships, categorical variables were analyzed using the Chi-squared test, while the Shapiro-Wilk test was applied to numerical variables to check normality and determine whether parametric or non-parametric tests were appropriate. The results show that demographic factors such as SeniorCitizen and Dependents, as well as service features including InternetService, OnlineSecurity, TechSupport, StreamingTV, and StreamingMovies, have a significant association with churn. In contrast, PhoneService does not show a meaningful impact. Company policy-related features such as Contract and PaymentMethod also strongly influence customer retention. Normality tests revealed that numerical variables do not follow a normal distribution at the 5% significance level, requiring non-parametric approaches. Hypothesis testing further indicated that there is insufficient statistical evidence to conclude that churned customers differ from retained customers in terms of tenure, MonthlyCharges, or TotalCharges.

Overall, the analysis highlights that churn is driven more by service quality, contract type, and customer demographics than by cost alone. These findings suggest that improving support services, offering flexible contracts, and tailoring engagement strategies for vulnerable customer groups can help reduce churn and strengthen customer loyalty.



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INTRODUCTION

Background of the study

This data set contains customer details of telecommunication company called "SpeedyCall" which is in USA. In this analysis we are interested to know whether a customer will churn or not according to customer behavior.

objectives

- To examine the impact of demographic variables on customer churn and use visualizations to analyze churn patterns across other key variables.
- To evaluate the impact of financial factors on customer churn.
- To analyze how service-related features influence customer churn (eg - PhoneService, InternetService, OnlineSecurity, TechSupport).
- To analyze how value-added service features influence customer churn (eg - StreamingTV, StreamingMovies).
- To evaluate how organizational policy-based variables impact customer satisfaction (eg - Contract, PaymentMethod, MonthlyCharges), which are some of the key variables related to company policies.

significance of the study

Understanding customer churn is crucial for telecommunication companies because acquiring new customers is significantly more expensive than retaining existing ones. A churn analysis helps the company identify patterns and key factors such as service quality, pricing, contract type, or customer support, that lead customers to discontinue their services. By studying churn, telecommunication companies can predict customer behavior, implement targeted retention strategies, improve customer satisfaction, and optimize business policies. Ultimately, churn analysis enables the company to reduce revenue loss, maintain a stable customer base, and gain a competitive advantage in a highly saturated market.

DATA

Focusing on the most relevant features allows for a clearer, more interpretable model that highlights actionable drivers of churn.

ABOUT VARIABLES

- **SeniorCitizen** – Whether the customer is a senior citizen or not (Categorical variable - 1, 0)
- **Dependents -** Whether the customer has dependents or not (Categorical variable - Yes, No)
- **tenure -** Number of months the customer has stayed with the company (measured in months - Numerical variable).
- **PhoneService -** Whether the customer has a phone service or not. (Categorical variable - Yes, No)
- **InternetService -** Customer's internet service provider. (Categorical variable - DSL,Fiber optic, No)
- **OnlineSecurity -** Whether the customer has online security or not. (Categorical variable - Yes, No, No internet service)
- **TechSupport -** Whether the customer has tech support or not. (Categorical variable - Yes, No, No internet service)
- **StreamingTV -** Whether the customer has streaming TV or not. (Categorical variable - Yes, No, No internet service)
- **StreamingMovies -** Whether the customer has streaming movies or not. (Categorical variable - Yes, No, No internet service)

- **Contract -** The contract term of the customer (Categorical variable - Month-to-month, One year, Two year)
- **PaymentMethod -** The customer's payment method (Categorical variable - Electronic check, Mailed check, Bank transfer (automatic), Credit card (automatic))
- **MonthlyCharges -** The amount charged to the customer monthly. (Numerical variable)
- **TotalCharges -** The total amount charged to the customer. (Numerical variable)
- **Churn -** This is the target variable that indicates whether the customer churned or not (Yes or No)

Data preprocessing

Customer dataset of Speedy Call company initially had 7043 observations and 21 variables. The dataset contains both numerical and categorical variables. Out of the 21 variables, 3 are numerical and the remaining are categorical (nominal or ordinal).

- Removed redundant or irrelevant variables to simplify the dataset and focus on key features. (removed variables - gender, Partner, MultipleLines, OnlineBackup, DeviceProtection, PaperlessBilling)

- In this dataset, there are 11 missing values in the TotalCharges variable. Therefore, records corresponding to those missing values are deleted. There are a very small number of missing values compared to the number of observations, they can be safely removed without requiring further investigation.

- Boxplots were generated for all numerical variables, and since no outliers were observed, no action is required.

Now cleaned data set have 7032 observations and 14 variables.

Data Analysis

Analysis categorical variables

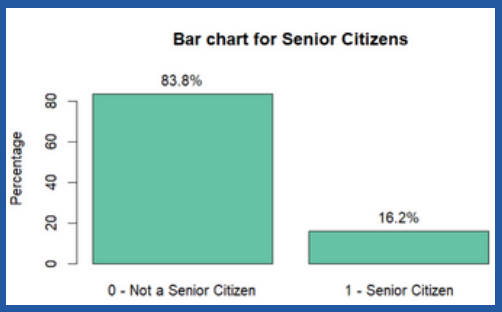

Figure 1: Bar chart for Senior Citizen

This bar chart (*Figure 1*) clearly shows that majority of individuals in the data set are not senior citizens.(In USA people who are older than 65 years are consider as senior citizens). Senior citizens may rely less on these services, resulting in their lower representation in the churn dataset. In addition, the percentage (17.3%) of senior citizens in the country is lower compared to that of the younger population.

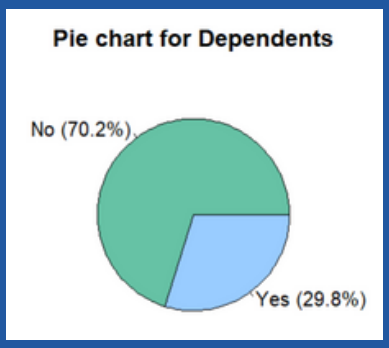

Figure 2 :Pie chart for Dependents

This pie chart (*Figure 2*) shows that 70% of customers do not have dependents. Therefore majority of individuals in this data set do not have obligation of a family.

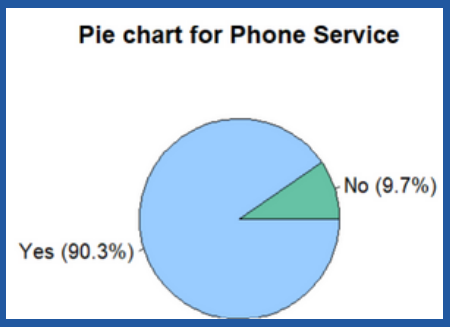

Figure 3 :Pie chart for Phone Service

This pie chart (*Figure 3*) indicates that the majority of customers of the company have the company's phone service, accounting for 90.32% of the customer base. This suggests that most customers rely on the phone service, which may be a key factor in analyzing usage patterns and churn behavior.

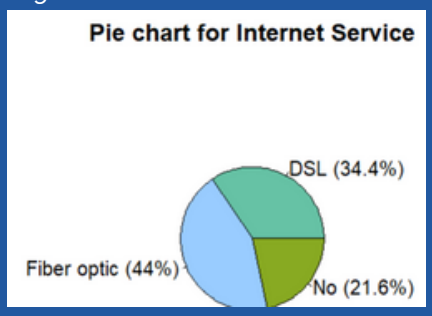

Figure 4 :Pie chart for Internet Service

The majority of customers rely on this company for internet service as well, with only 21.6% of customers not having an internet connection from Speedy Call. So, customer churn may be associated with this variable.

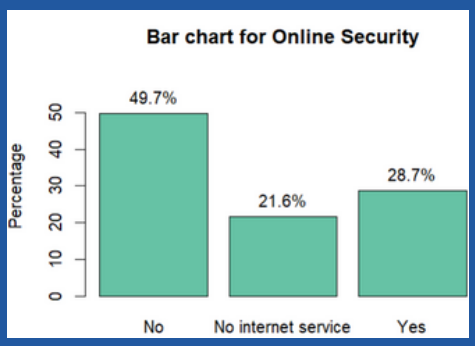

Figure 5:Bar chart for Online Security

Above bar graph (*Figure 5*) clearly depicts that the majority of customers(49.7%) have not subscribed to the online security service.

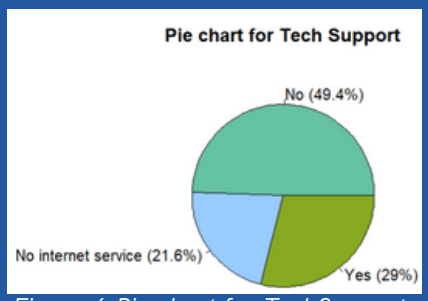

Figure 6:Pie chart for TechSupport

78.4% of customers receive their internet service from Speedy Call. However, among those with an internet connection, 49.4% have not subscribed to the company's Tech Support service.

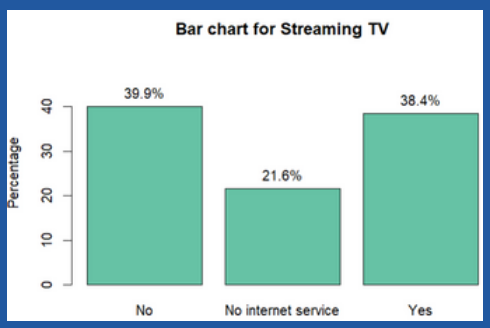

Figure 7:Bar chart for Streaming TV

Above bar chart (*Figure 7*) clearly shows that the majority of customers do not have Streaming TV service, as well as internet connection (since streaming TV is not possible without an internet connection).

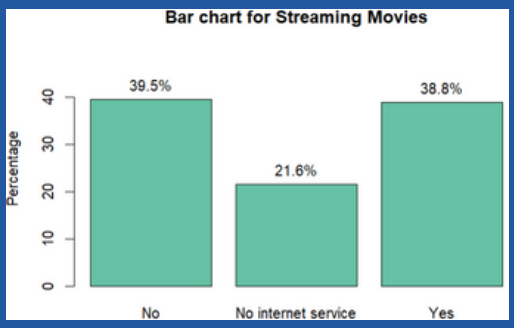

Figure 8:Bar chart for Streaming Movies

Bar chart for StreamingMovies (*Figure 8*) has the same characteristics as StreamingTV. If we were to fit a logistic regression model, this situation may cause redundancy. However, in our analysis, both should be kept since they are important services for the company's customers. Improving these value-added services may help reduce customer churn.

Figure 9:Pie chart for Contract

Figure 9 depicts that majority of customers (55.1%) are on month-to-month contracts, followed by two-year contracts at 24%, and one-year contracts at 20.9%. This suggests that most customers prefer short-term flexibility, which may correlate with higher churn risk.

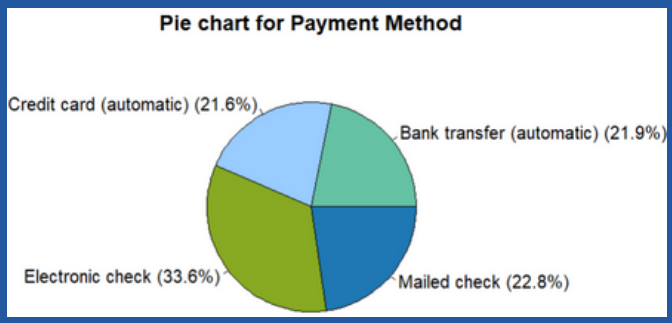

Figure 10 :Pie chart for Payment Method

Figure 10 shows that about one-third of customers use electronic check the most, while the other payment methods are used by customers in almost equal amounts.

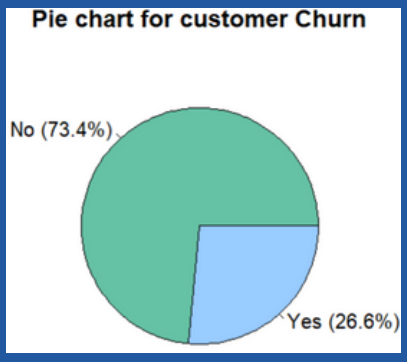

Figure 11 :Pie chart for churn

Figure 11 shows that most customers stayed, but more than a quarter left, which is important for the company to address.

Analysis of numerical variables

Monthly Charges

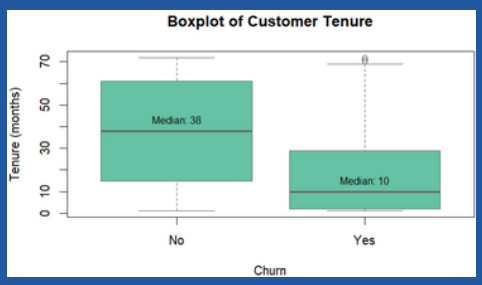

Figure 12 :Boxplot for Customer Tenure categorized by Churn

These two boxplots clearly show that the median tenure of churned customers is significantly lower than that of retained customers. Therefore, customers are more likely to churn in the early months of service. There are some outliers in "Yes" category and that category is right skewed.

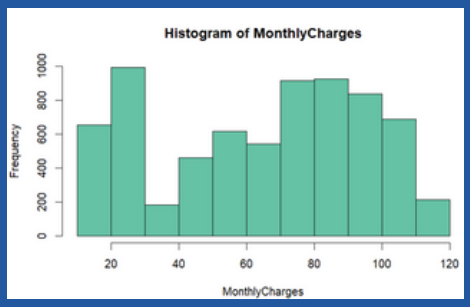

Figure 13 :Histogram for Monthly Charges

```
        Shapiro-Wilk normality test

data:  sample_data
W = 0.92191, p-value < 2.2e-16
```

Figure 14 :Output of Shapiro test

Both the histogram (*Figure 13*) and the Shapiro-Wilk test (*Figure 14*) indicate that Monthly Charges do not follow a normal distribution. (Since R does not support the Shapiro test for samples larger than 5000, the test was performed on a random sample of 5000 observations).

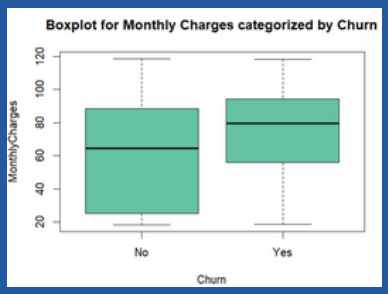

Figure 15:Boxplot for Monthly Charges categorized by Churn

The boxplot (*Figure 15*) for Monthly Charges by churn category shows that churned customers have a higher median monthly charge than retained customers. This suggests that higher monthly charges may be a reason for customer churn.

Plots for Total Charges

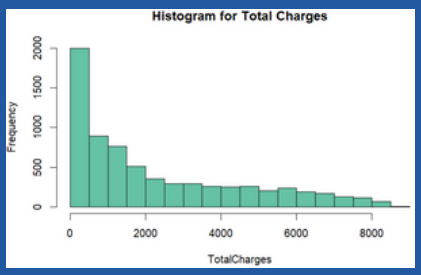

Figure 16:Histogram for Total Charges

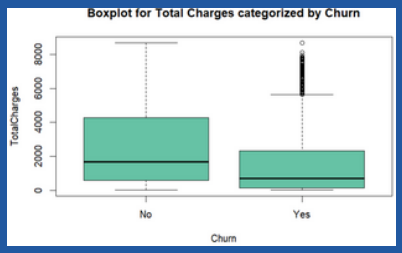

Figure 17:Boxplot for Total Charges categorized by churn

Histogram for Total Charges (*Figure 16*) clearly depicts that the variable does not follow a normal distribution, showing a right-skewed pattern. The boxplots show that the median Total Charges of retained customers is higher than that of churned customers. This is likely because customers with higher tenure tend to have lower tendency to churn.

Hypothesis tests and Interpretations

Evaluating Demographic Influences on Customer Churn Using the Chi-Squared Test

```
        Pearson's Chi-squared test with Yates' continuity correction

data:  t_10
X-squared = 158.44, df = 1, p-value < 2.2e-16
```

Figure 18:Senior Citizen(Out put of chi squared test)

```
        Pearson's Chi-squared test with Yates' continuity correction

data:  t_11
X-squared = 186.32, df = 1, p-value < 2.2e-16
```

Figure 19 :Dependents(Output of chi squared test)

H_0 : No relationship between the variables.
H_1 : There is a relationship between the variables.

Rejection criteria: if p < alpha then reject null hypothesis.
let significance level = 5%

So, 2.2e-16 <0.05 therefore we have enough evidence to reject null hypothesis at 5% level of significance. Therefore, we can conclude that both Senior Citizen and Dependents have significant association with customer churn.

Understanding Customer Churn Through Service Subscription Patterns: A Chi-Square Approach

```
        Pearson's Chi-squared test with Yates' continuity correction

data:  t_12
X-squared = 0.87373, df = 1, p-value = 0.3499
```

Figure 20: phone service (output of chi square test)

```
        Pearson's Chi-squared test

data:  t_13
X-squared = 728.7, df = 2, p-value < 2.2e-16
```

Figure 21: Internet service (output of chi squared test)

```
        Pearson's Chi-squared test

data:  t_14
X-squared = 846.68, df = 2, p-value < 2.2e-16
```

Figure 22: Online Security (output of chi squared test)

```
        Pearson's Chi-squared test

data:  t_15
X-squared = 824.93, df = 2, p-value < 2.2e-16
```

Figure 23:Tech Support (Chi squared output)

```
        Pearson's Chi-squared test

data:  t_16
X-squared = 372.46, df = 2, p-value < 2.2e-16
```

Figure 24: Streaming TV(output of chi squared test)

```
        Pearson's Chi-squared test

data:  t_17
X-squared = 374.27, df = 2, p-value < 2.2e-16
```

Figure 25: Streaming Movies (output of chi squared test)

Using above figures (20, 21, 22, 23, 24, 25) we can conclude that only Phone service(*Figure 20*) is not significant, all the other variables (*Figure 21, 22, 23, 24 and 25*) shows a significant association with churn at 5% level of significance.

Checking Churn Link with Company Policy Features Using Chi-Squared Test

```
        Pearson's Chi-squared test

data:  t_21
X-squared = 1179.5, df = 2, p-value < 2.2e-16
```

Figure 26: Contract (chi squared test output)

```
        Pearson's Chi-squared test

data:  t_22
X-squared = 645.43, df = 3, p-value < 2.2e-16
```

Figure 27: Payment Method (chi squared test output)

All above p- values are less than 0.05, therefore we have enough evidence to conclude that there is significant association between above each variable and customer churn.

Contract and PaymentMethod show a significant association with churn, indicating that company policy-related features can strongly influence customer retention.

Hypothesis tests regarding numerical variables

```
        Shapiro-Wilk normality test

data:  sample_data
W = 0.90562, p-value < 2.2e-16
```

Figure 29: Tenure (output of Shapiro test)

```
        Shapiro-Wilk normality test

data:  sample_data
W = 0.92364, p-value < 2.2e-16
```

Figure 30: Monthly Charges (output of Shapiro test)

```
        Shapiro-Wilk normality test

data:  sample_data
W = 0.86203, p-value < 2.2e-16
```

Figure 31: Total charges (output of Shapiro test)

Above three figures (figure 26,27 and 28) shows that all numerical variables do not follow a normal distribution at 5% level of significance. Therefore we need to use non parametric-tests.

Tenure-Based Churn Analysis: Non-Parametric Approach

H_0: Median tenure of churned customers ≥ median tenure of retained customers
H_1: Median tenure of churned customers < median tenure of retained customers

```
        Wilcoxon rank sum test with continuity correction

data:  tenure by Churn
W = 7154668, p-value = 1
alternative hypothesis: true location shift is less than 0
```

Figure 32: Median Tenure vs. Churn (Wilcoxon Test)

Significance level - 5%
Rejection criteria: If U <= critical value then reject H_0 or If p <= alpha then reject H_0

p = 1 > 0.05 therefore we do not reject null hypothesis at 5% level of significance.
We do not have sufficient statistical evidence to support the claim that churned customers have shorter tenure than retained customers.

Monthly Charges - Based Churn Analysis: Non-Parametric Approach

H_0: Median Monthly Charges of churned customers ≤ median of retained customers.
H_1: Median Monthly Charges of churned customers > median of retained customers.

```
        Wilcoxon rank sum test with continuity correction

data:  MonthlyCharges by Churn
W = 3663499, p-value = 1
alternative hypothesis: true location shift is greater than 0
```

Figure 33: Monthly Charges Vs. Churn (Wilcoxon Test)

p = 1 > 0.05, therefore, we do not have enough evidence to reject the null hypothesis at the 5% level of significance. Hence, we conclude that we cannot confirm churned customers have higher Monthly Charges than retained customers.

Total Charges - Based Churn Analysis: Non parametric Approach

H_0: Median Total Charges of churned customers ≥ median Total Charges of retained customers.
H_1: Median Total Charges of churned customers < median Total Charges of retained customers.

```
        Wilcoxon rank sum test with continuity correction

data:  TotalCharges by Churn
W = 6288982, p-value = 1
alternative hypothesis: true location shift is less than 0
```

Figure 34: Total Charges Vs Churn(Wilcoxon Test)

p = 1 > 0.05, therefore, we do not have enough evidence to reject null hypothesis at 5% level of significance. Therefore, we do not have enough evidence to conclude that churned customers have lower median Total Charges.

General Discussion and Conclusion

- Demographic variables such as SeniorCitizen and Dependents show a significant association with customer churn, indicating that age-related status and family responsibilities may influence churn behavior.
- Financial factors like Monthly Charges and Total Charges showed no significant difference between churned and retained customers, suggesting cost alone may not drive churn.
- Service features like InternetService, OnlineSecurity, and TechSupport show a strong link to churn, unlike PhoneService, which has no significant impact, possibly because most customers have phone service by default, making it less influential in churn decisions.
- StreamingTV and StreamingMovies show a significant association with churn, suggesting that even small differences in how customers use these value-added services can influence whether they stay or leave.
- Customers' decisions to stay or leave seem to depend not just on how much they pay, but also on how they pay and the type of contract they have. Since Contract and PaymentMethod show a strong link to churn, it's clear that company rules around billing and plans can play a big role in keeping customers or losing them.

To reduce customer churn, the company should focus on improving services and making contracts more flexible. Since one out of four customers has already left, it's important to give extra support to groups like senior citizens and families with dependents, and to strengthen services such as OnlineSecurity, TechSupport, and streaming options. Clear and flexible contracts, along with easier payment methods, can also help customers feel more satisfied and stay longer.